Name: ____________________

Class: ____________________

Date: ____________________

Mr. Rawson • GCSE Physics

MINI REVIEW 01 — Distance and Speed

Optional. Not marked by me, not recorded anywhere. We mark it together next week.

Where this fits the January mock. The mock covers Forces, Energy, Electricity, Particle and Atomic. This is the **first slice of Forces**. Packets 1–14 walk forward through Year 10 in taught order, so by the winter break you hold the whole of it in one folder.

Key concept

DISTANCE vs DISPLACEMENT (the difference is direction)

- **Distance** — how far you moved, added up along the whole path. No direction. **Scalar**.
- **Displacement** — the straight line from start to finish, *and* the direction of that line. **Vector**.

Walk 400 m right round a running track and stop where you started: distance **400 m**, displacement **0**.

Key concept

SPEED (scalar — no direction)

$$\text{distance} = \text{speed} \times \text{time} \qquad s = vt$$

s in metres (m) · v in metres per second (m/s) · t in seconds (s)

Real speed is hardly ever constant — walking, running and driving all change speed constantly. When it does change, $s = vt$ with the **average** speed still works over the whole journey.

Typical speeds you are expected to recall:

walking	~1.5 m/s	running	~3 m/s
cycling	~6 m/s	sound in air	~330 m/s

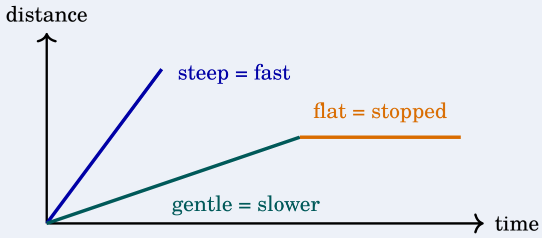
Worked example. A cyclist travels 1500 m in 250 s. Find the average speed.

Rearrange $s = vt$ to $v = \frac{s}{t} = \frac{1500}{250} = \mathbf{6\ m/s}$ — which is the typical cycling speed above, so the answer is believable. Always ask whether your number is a sensible speed for the thing that is moving.

Key concept

DISTANCE–TIME GRAPHS (distance up, time across)

The gradient is the speed. Steeper line, faster object.



A **flat** line means the object is **not moving** — time passes, distance does not change. It does *not* mean it has gone back to the start.

Name: ______________________________ **Date:** ______________________

MINI REVIEW 01 — the core

Ten items, about ten minutes. Write something for each one even if you are not sure.

1. State whether each quantity is a **scalar** or a **vector**.

   distance _____________ displacement _____________ speed _____________

2. A runner completes one full lap of a 400 m track.

   Distance travelled: _____________ Displacement: _____________

3. Write down the equation linking distance, speed and time, in symbols.

   ____

4. A car travels at a steady 25 m/s for 40 s. How far does it go?

   ____

5. A walker covers 900 m in 600 s. Calculate the average speed.

   ____

6. Give the typical speed of each, in m/s.

   walking ___________ running ___________

   cycling ___________ sound in air ___________

7. On a distance–time graph, what does the **gradient** of the line tell you?

   ____

8. On a distance–time graph, what does a **horizontal (flat)** line mean?

   ____

9. Two straight lines are drawn on the same distance–time graph. Line A is steeper than line B. Which object is moving faster, and how do you know?

   ____

   ____

10. Thunder is heard 6 s after the lightning is seen. Taking the speed of sound in air as 330 m/s, how far away was the strike?

    ____

Stretch — not marked in class. A student walks 30 m north, then 40 m east, in a total of 50 s. **(a)** What total *distance* did she walk? **(b)** What was her *displacement*? **(c)** What was her average *speed*?